ISM Note 2

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1. Spiral galaxies (continued)

The Toomre's Q parameter is a very important parameter to analyze the stability of the disk, and it is defined by

$$Q = \frac{c_s \kappa}{\pi G \Sigma}$$

where c_s is the sound speed, Σ is the surface density, κ is the epicyclic frequency, which is defined by

$$\kappa = \sqrt{\frac{2\Omega}{r}\frac{d\Omega}{dr}}$$

If $Q < 1$, the disk is **gravitationally unstable**, and if $Q > 1$, the disk is gravitationally stable, while if $Q \approx 1$, the disk is marginally stable.

The **unstable** here means the gas will fragment into clumps and then **collapse to form stars**. So from the definition of Toomre's Q, we can see that when the thermal pressure is inenough to support the gas against gravity, the disk will be unstable and thus collapse. This physical picture is intuitively reasonable and from that we can understand how the Toomre's Q works and why it is important.

Give an example here. From the previous lecture, we know the gas in the spirals mainly flow along the spiral arms, and since the spiral arms are essentially the density waves that propagate through the galactic disk, the gas will be compressed and this will directly increase the surface density Σ, and thus the instability condition $Q < 1$ can be easily satisfied. Therefore, the spiral arms are the ideal place for star formation, and this is consistent with the observation that we can see many young stars in the spiral arms.

Note that the Toomre's Q is not the only criterion to analyze the stability of the disk, for example, if $Q > 1$, the disk is either linearly stable or nonlinearly unstable. From the formulation of the Toomre's Q, we can see that thermal pressure + rotation can stabilize the disk, and sometimes the $\mathbf{v}_{\mathrm{turb}}$ can replace the sound speed $\mathbf{c}_{\mathbf{s}}$, which means the turbulence can also stabilize the disk, in other words, **the turbulence can support the gas against collapse**. To give an intuitive picture, the turbulence to some extent can describe the random motion of the gas, and similarly this **random motion** is literally what the thermal pressure means.

2. Elliptical galaxies

The elliptical galaxies are almost spherical, and NFW profile is a good description for the density distribution of the elliptical galaxies. **The Navarro–Frenk–White (NFW) profile** is one of the most commonly used models for the structure of dark matter halos. It describes the **average density** of dark matter in galaxy and galaxy cluster halos as **a function of distance** from their center.

The formula for the NFW profile is given by

$$\rho(r) = \frac{\rho_0}{\frac{r}{r_s}(1 + \frac{r}{r_s})^2}$$

where ρ_0 and r_s are two parameters, here r_s is the core radius and is usually about **1-2 kpc**. The NFW profile is a cuspy profile, which means the density increases sharply as r decreases.

The below figure shows the NFW profile

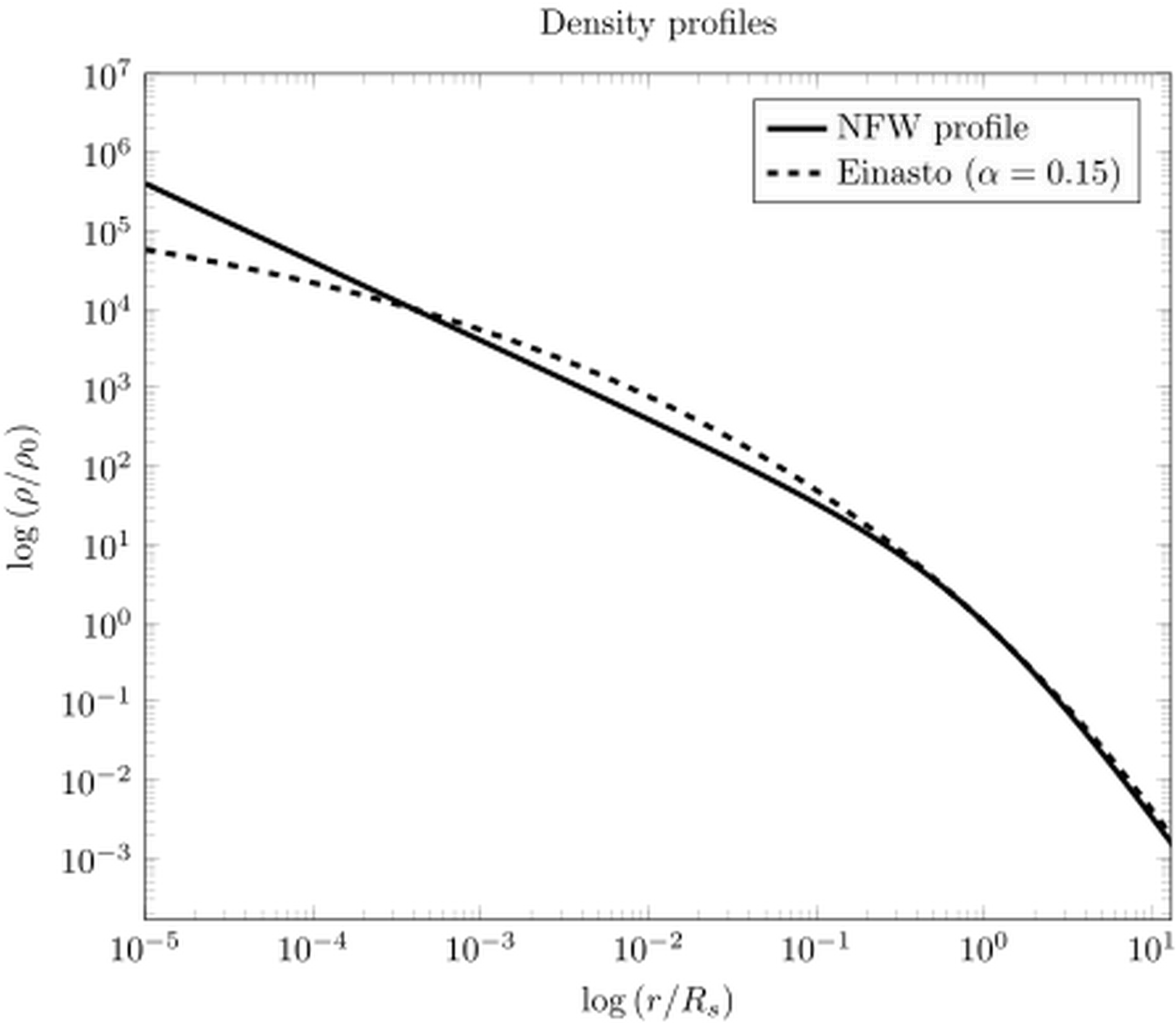

Fig 1. The NFW profile.

The spherical shape of the elliptical galaxies means **they are not supported by rotation**, since rotation will break the spherical symmetry which is inconsistent with the spherical shape of ellipticals. So the question here is: what supports the elliptical galaxies against collapse? The answer is **the random motion** or **"heat"**. Quantitatively

$$v_{\text{rot}} \leq 0.1\sigma$$

which means the rotation velocity is much smaller than the random motion. So for an elliptical galaxy, **the temperature must be much higher than the spirals** to support the galaxy against collapse.

Since the elliptical galaxies are spherical, we can not abstract them as a 2d object as what we do for spirals, but only consider them as **3d objects**. We can write down the hydrostatic balance equation for the elliptical galaxies, which is given by

$$\frac{d(\rho\sigma^2)}{dr} = -\frac{GM(r)\rho}{r^2}$$

This equation is not hard to derive from Newtonian mechanics. By the way, if we replace σ with $v_{\rm rot}$, then we can get the turbulence equation.

A natural question is: Why spirals and ellipticals are so different? This question is hard to answer, but we can have several clues for this. The first clue is because of **early star formation vs late star formation**. The ellipticals are the early-types, while the spirals are the late-types.

What do we mean by "early-types" and "late-types"? Historically, in the Hubble sequence, elliptical galaxies were called **"early-types"** and spiral galaxies were called **"late-types"**. This naming was based on their morphology in the "tuning fork" diagram and was once thought to represent an evolutionary path, but this is now known to be incorrect.

Back to our discussion, the early star formation can release a large amount of energy, which can heat the gas and then get the random motion stronger, thus the galaxy can be supported by the random motion more than rotation, and then the galaxy will be more likely to be an elliptical.

Interestingly, **the spirals can transform into ellipticals**, and the physical process for this transformation can be described as follows. When two spirals merge together, the angular momentum can be destroyed, and the star formation can be triggered, which can heat the gas and thus make the galaxy an elliptical.

The following figure shows the spiral mergers.

Fig 2. Spiral mergers

The second clue is the **environmental effects**. For example, galaxy can be affected by the mergers. The mergers can destroy the angular momentum and cause star burst, thus transforming the galaxy from a spiral to an elliptical.

3. Star-forming galaxies vs quiescent (passive) galaxies

Except for the classification based on morphology, we can also divide the galaxies into

star-forming galaxies and **quiescent (passive) galaxies**. This classification is based on the star formation rate (SFR) of the galaxies $\dot{M}_*$, which measures the rate of newly formed star mass.

We have another quantity named the **specific star formation rate (sSFR)**

$$\mathrm{sSFR} = \frac{\dot{M}_*}{M_*}$$

which is a good indicator to distinguish the star-forming galaxies and quiescent galaxies, since the star formation rate can depend on the galaxy's mass and may not be a good estimate for how "active" the galaxy is.

The star-forming galaxies have a high sSFR, which is typically about $\geq 10^{-11}\mathrm{yr}^{-1}$. As an example, for our Milky Way, it is star-forming, and its sSFR is about $10^{-11}\mathrm{yr}^{-1}$.

For active galaxies we usually say it is **blue**, while for quiescent galaxies we usually say it is **red**. Interestingly, though most spirals are star-forming, a population of **"red spirals"** exists. One possible reason for this might be that the gas is heated by the AGN feedback, thus without enough cold gas the star formation is quenched, and the galaxy becomes quiescent or red.

active galaxy

The "active" here does not mean star-forming, but **small, highly energetic, non-stellar events within the galaxy**. These events are usually in the radio band.

One example of the active galaxy is the **Seyfert galaxy**, which is a spiral having bright and compact nuclei which cannot be explained by stars, so there must be some highly energetic and non-stellar events within the Seyfert galaxy.

Another example is the **radio galaxy**, which has double radio lobes. The lobes can extend to a very large scale and often many times larger than the galaxy itself. No process related to individual stars can produce such large scale and powerful radio emission. Besides, the lobes are fed by two highly focused, narrow jets of plasma that are ejected in opposite directions from the galaxy's center, but stars and supernovae explode are more or less spherical and cannot produce such narrow jets. So the radio galaxy is another example of the active galaxy.

One image of the radio galaxy is shown as follows.

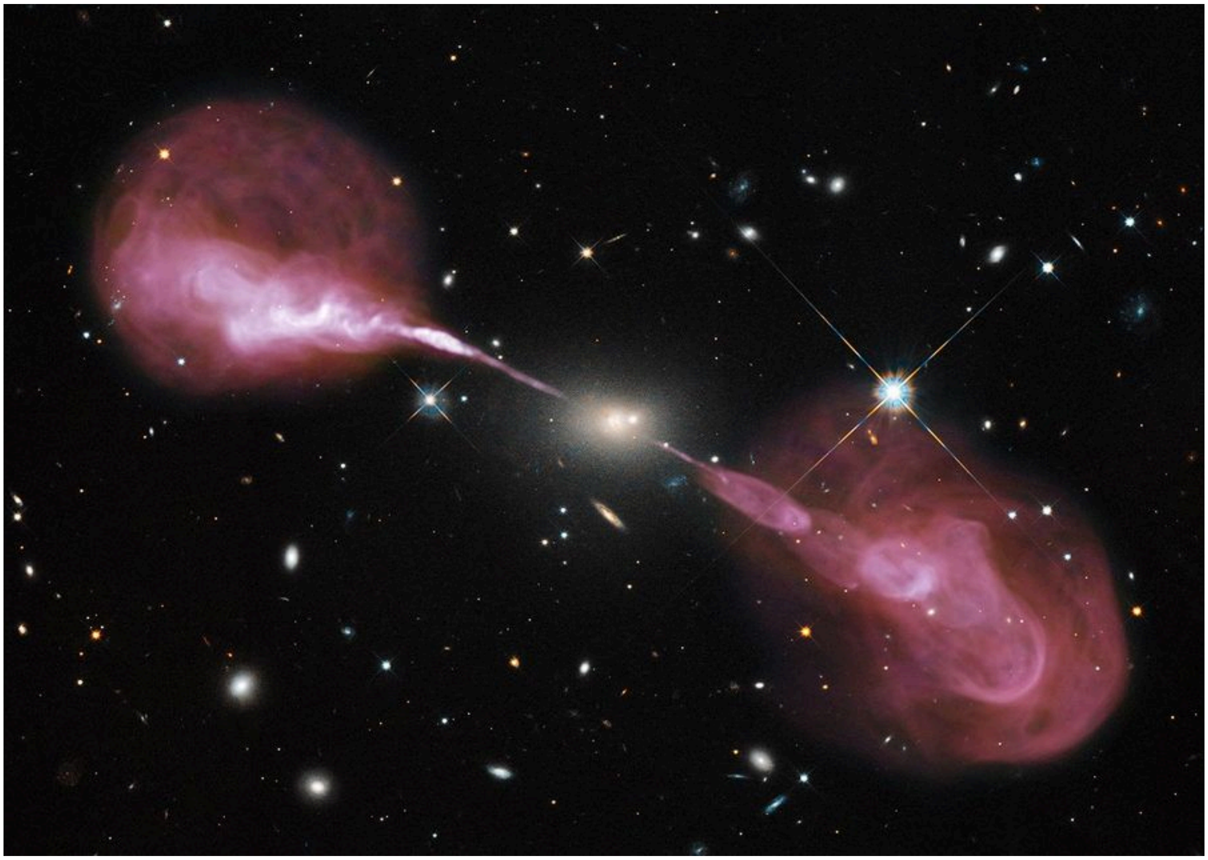

Fig 3. The radio galaxy

Beside for these, there is another active galaxies named **quasars**, or "quasi-stellar" objects, which are very bright. All these active galaxies can be explained by AGN, namely active galactic nuclei. But here we will not go into the details of AGN.

4. Stellar dynamics

Now we turn to the math and start from the distribution function

$$f(\mathbf{x}, \mathbf{v}, t) = \frac{dN}{d^3x d^3v}$$

which describes the number of stars in the phase space. Then naturally we want to study **the evolution of the distribution function**, and the equation governing the evolution of the distribution function is called the **Vlasov equation**, which is given by

$$\frac{\partial f}{\partial t}+\frac{\partial \mathbf{x}}{\partial t}\cdot\frac{\partial f}{\partial \mathbf{x}}+\frac{\partial \mathbf{v}}{\partial t}\cdot\frac{\partial f}{\partial \mathbf{v}}=0$$

And

$$\frac{\partial \mathbf{x}}{\partial t}+\mathbf{v}\cdot\nabla_{\mathbf{x}} f+\mathbf{g}\cdot\nabla_{\mathbf{v}} f=0$$

where $\mathbf{g}=-\nabla\Phi$ is the gravitational acceleration and Φ is the gravitational potential.

Note that we are in a **6d phase space** which is difficult to deal with, so we can integrate the distribution function over the velocity space first and get

$$\int\frac{\partial f}{\partial t}d^3v+\int\mathbf{v}\cdot\nabla_{\mathbf{x}} f d^3v+\int\mathbf{g}\cdot\nabla_{\mathbf{v}} f d^3v=0$$

The first term is

$$\frac{\partial}{\partial t}\int f d^3v=\frac{\partial n}{\partial t}$$

which is the time derivative of the number density. The second term is

$$\nabla_{\mathbf{x}}\cdot\int\mathbf{v} f d^3v=\nabla_{\mathbf{x}}\cdot(n\langle\mathbf{v}\rangle)$$

where

$$\langle\mathbf{v}\rangle=\frac{\int\mathbf{v} f d^3v}{\int f d^3v}$$

is the mean velocity. The third term is

$$\int\mathbf{g}\cdot\nabla_{\mathbf{v}} f d^3v=0$$

because of the boundary condition that f goes to zero as v goes to infinity.

And combining all these together, we can get the **continuity equation**

$$\frac{\partial n}{\partial t} + \nabla_{\mathbf{x}} \cdot (n\langle \mathbf{v} \rangle) = 0$$